Have you made your AirSpec?

An open-source specification for crew rules



AirSpec

AVIATION INTERPRETED RULES SPECIFICATION

Viktor Forsman
Maz Lakadia

Instructions:

"Buy a bottle of milk. If they have eggs, get 12."

At the store, you see they have eggs. What do you buy?

Option A
1L milk
12 eggs

Option B
12 x 1L milk

Who are we?



Viktor Forsman

- MSc in Mathematical Optimization
- 15 years in crew optimization, including Jeppesen, Cathay Pacific and VeeOne
- Current focus on crew optimization and AI



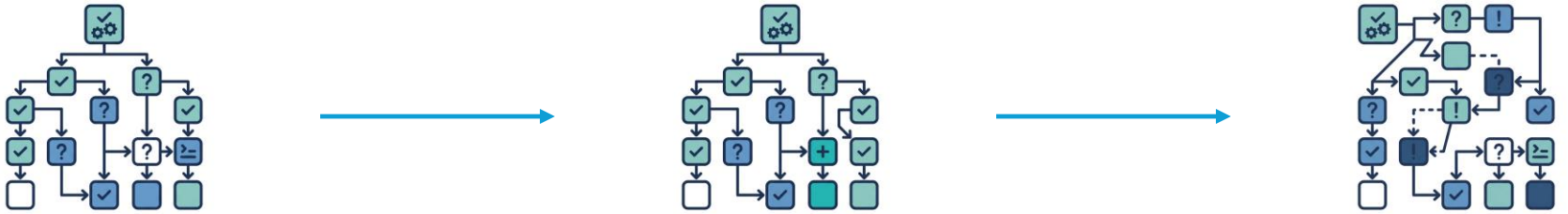
Maz Lakadia

- Background in OR and software
- A decade in flight operations, including Merlot and CAE
- Experience in writing large rule sets across clients and systems (crew tracking, planning, training, etc)

What do we keep seeing?

What do we keep seeing?

Rules re-implemented across systems



Pairing Rostering Tracking

Catalogue, interpret, implement and document again

Repeated effort. Long projects. High cost. High Risk

What do we keep seeing?

Rules re-implemented across fleets and crew groups

Same underlying rule



Fleet A

Unintended differences

Compliance risk

Unnecessary restrictions

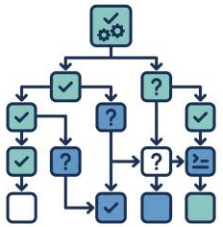


Fleet B

Different roster patterns. Less flexibility to fly together.

What do we keep seeing?

Rules re-implemented across teams and departments



Crew Management Systems

?

Hard to integrate



Planning



Payroll



Training



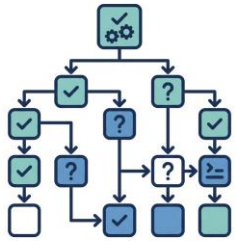
Manual processes

Other teams uses rough approximations of crew rules

Popular solutions

Popular solutions

Keep implementation and documentation in sync



Implementation

Out of date

or too vague

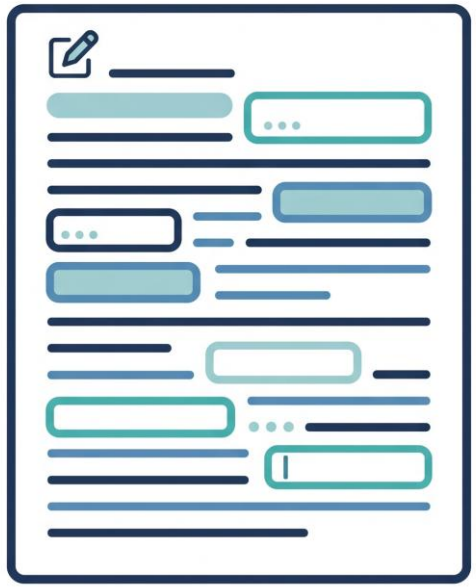


Documentation

Back to interpreting the rule

Popular solutions

“Natural language” rule configuration



Definitions remain ambiguous

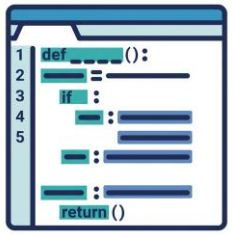
Popular solutions

Document to code convertor



Rule text

Lots of “degrees of freedom”



Rule code

Low cross-system reuse



AirSpec

Aviation Interpreted Rules Specification

AirSpec

A specification we develop together



Free and open source. Forever.

AirSpec

Rule knowledge in a formal structure



Rule text

Interpretation

Assumptions and context

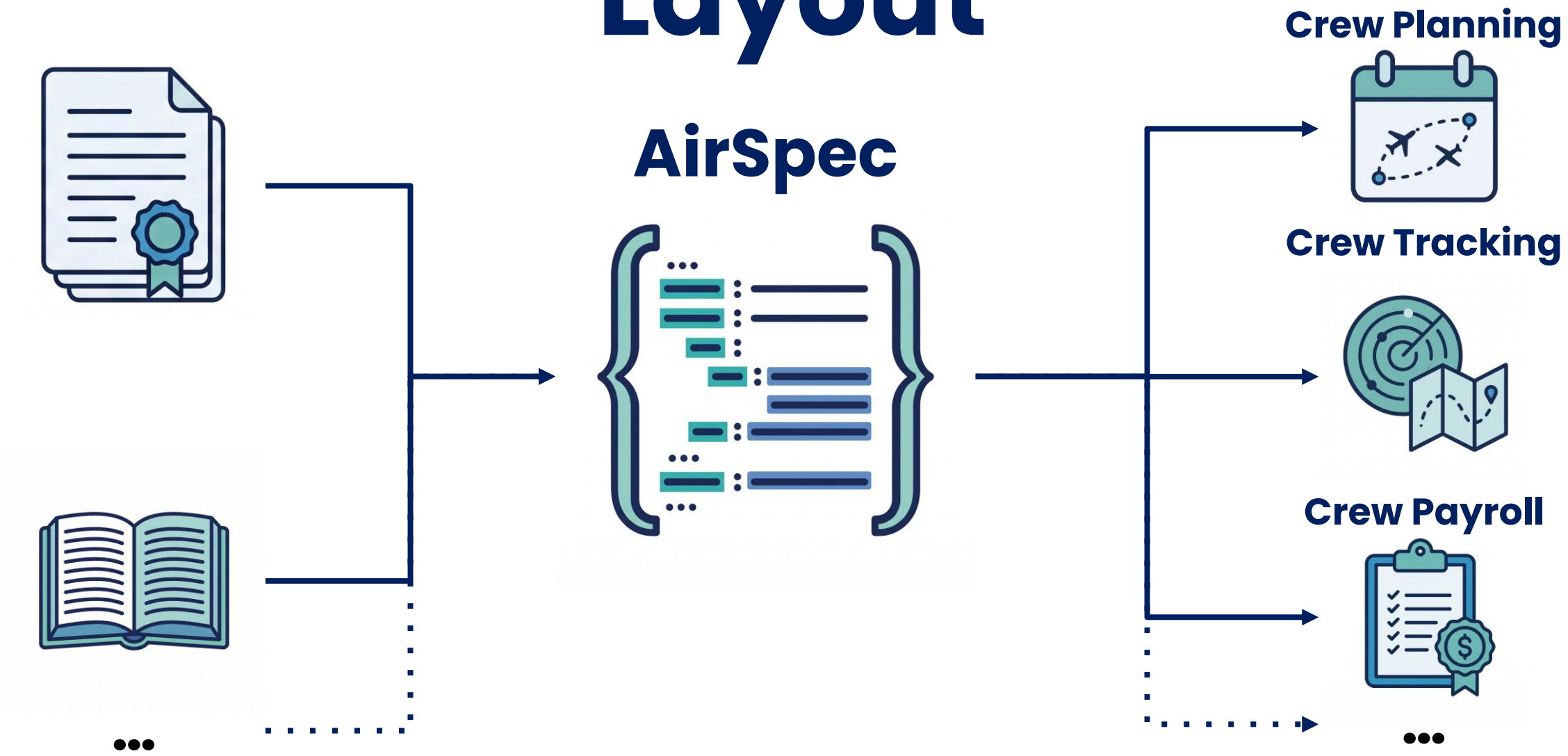


Machine-readable

Make ambiguity visible

Check and validate structure, consistency and duplication

Layout



Why now?

Tools are cheaper to build

Tools are cheaper to build

Shared specification + **AI-assisted development**

A common foundation

More teams can build on it

Airlines convert what they need. Vendors add support.

The community builds on each other’s work

What does this enable?

Catch questions before implementation



Source documents

Guided interpretation

People + tools



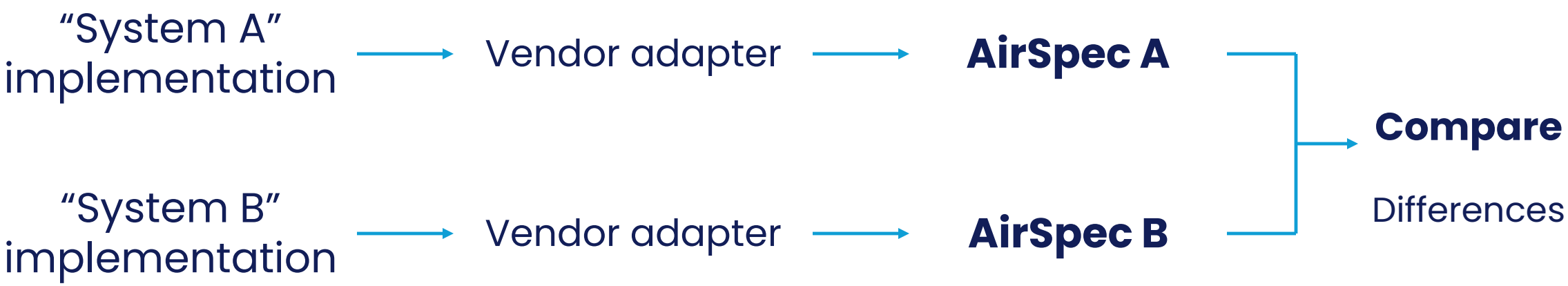
AirSpec

Ambiguities / Duplication / Missing assumptions

Resolve questions before they become code

What does this enable?

Compare existing implementations



Make differences in interpretation visible

What does this enable?

Understand a proposed rule change

Old AirSpec

New AirSpec

→ **Difference + example generator** →

Before / after cases

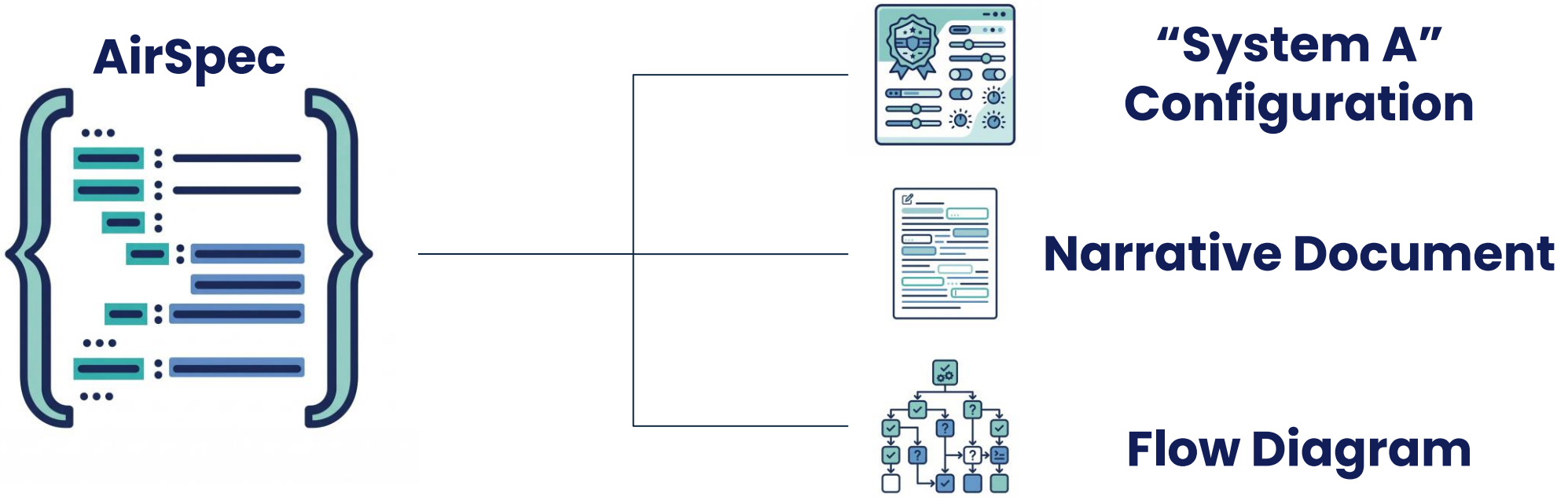
What changes?

Which boundary cases need testing?

Review the meaning of a change before implementation

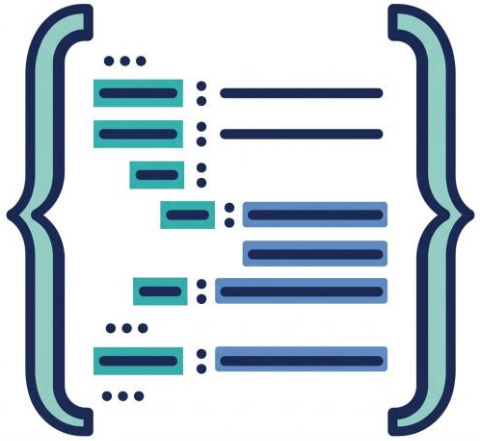
What does this enable?

Convert your AirSpec to the view you need



Automate creation of rule configuration, documents, and diagrams

Where are we now?



AirSpec V0 is available for public viewing and comment



Community groups setup on LinkedIn and Github

Help shape AirSpec

Not convinced? Tell us why.

Good idea? Stay in the loop.

Interested in implementing or getting involved? Let us know.

https://www.linkedin.com/groups/42821328/
https://github.com/airspec-aero/airspec

Open the feedback form